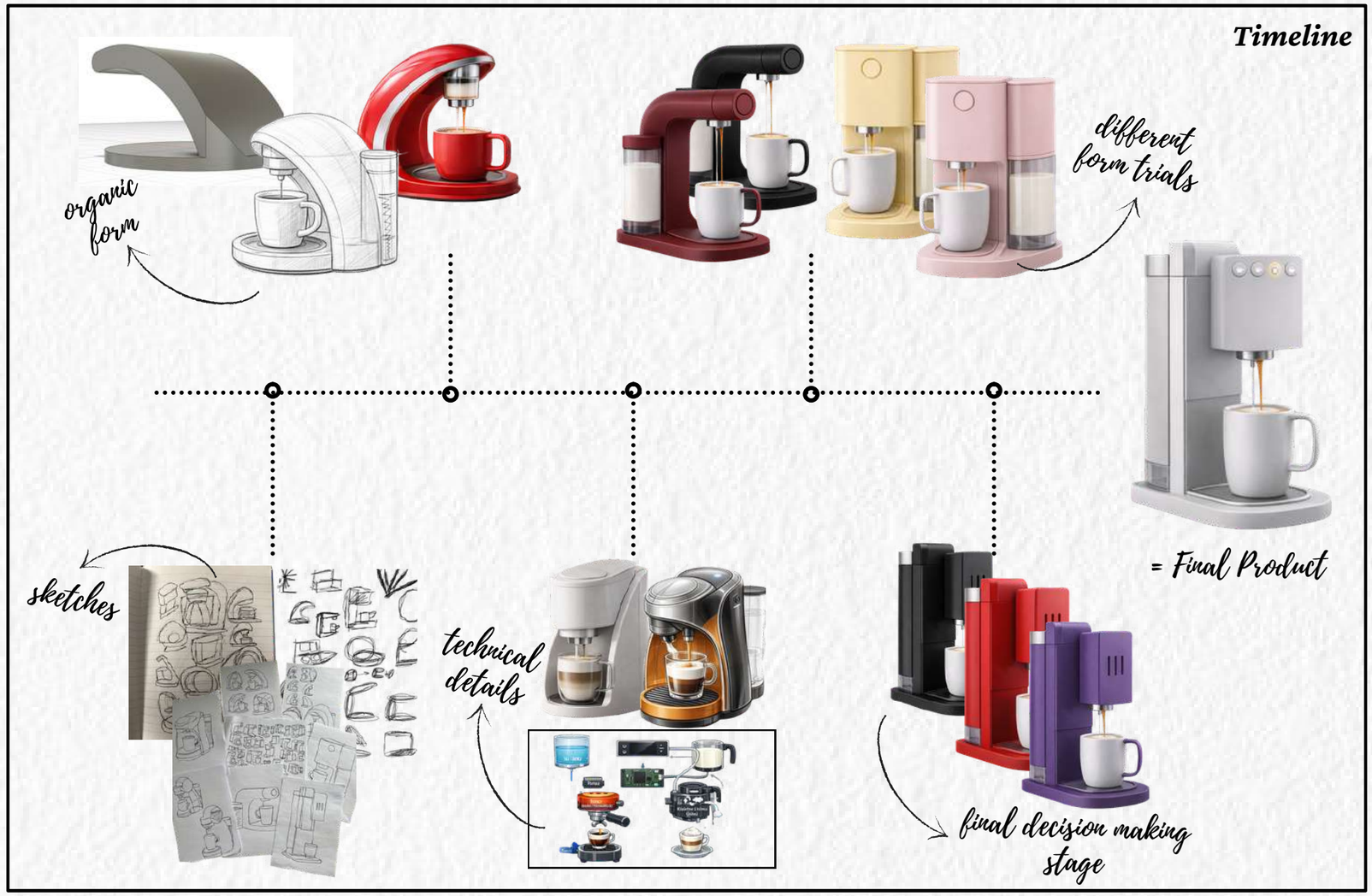
DESIGN MILESTONES



Keywords

- **Compact appliance**
- **User friendly interface**
- **Fast and easy preparation**
- **Clear feedback**
- **Home coffee experience**
- **Minimal design**

Impactful Prompts

Prompt 1
I am designing a compact milk-based coffee machine for home use. Transform my hand sketch into a realistic 3D product render while keeping the main proportions and form language. The design should look minimal, modern and suitable for small kitchens.

Prompt 2
Analyze my coffee machine concept and provide feedback on usability, especially focusing on the control interface, button placement and how users interact with the machine during the coffee preparation process.

AI Response Summaries

Sketch Visualization → AI converted rough sketches into realistic product renderings.
Interface Suggestions → AI proposed simple button layouts and display integration.
Research Support → AI helped gather information about milk-based coffee machines and their interface structures.

What I Accepted / Rejected / Transformed

Accepted
- AI visualizations that helped present my sketches more clearly
- Some form variations that helped me see the product from different perspectives

Rejected
- Interface suggestions that made the product look too complicated
- Form variations that were too bulky for a home coffee machine

Transformed
- My hand sketches were turned into clearer product visuals with the help of AI
- The surface lines on the machine were transformed into functional coffee selection buttons

Design Decisions

During the design process, I focused on creating a compact coffee machine suitable for home use. It was important for me that the product would be easy, quick and practical to use, so I simplified the interface by using small coffee selection buttons. I also removed unnecessary surface lines to make the product look cleaner and more minimal while keeping the interaction clear and understandable.

My Reflection

Using AI tools helped me visualize my sketches faster and see different possibilities for the product form. It was useful for turning rough sketches into clearer visuals and exploring small changes in the design. However, AI mostly acted as a supporting tool in the process, and the main design direction and final decisions were made by me.

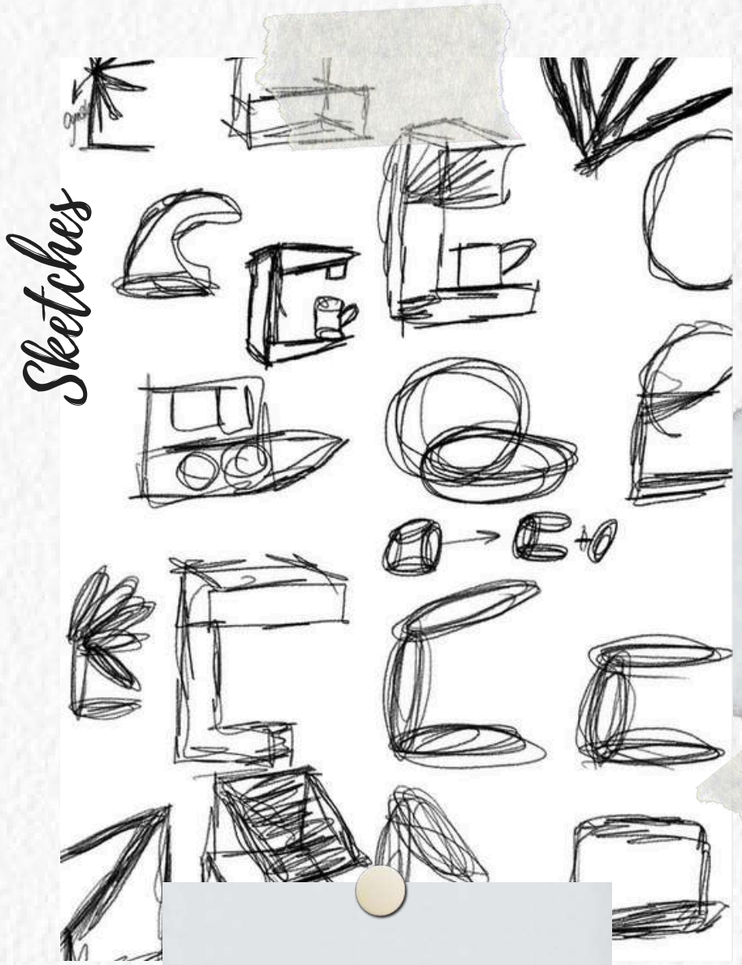
AI Role Tags

Clarifier → Helped clarify the design direction and interface ideas.
Re-framer → Suggested alternative interpretations of the sketch and product form.
Expander → Expanded the sketch into different visual product variations.
Critique → Helped evaluate the usability of the interface and control layout.

Lara Coşkun - S034269

CONSEPT BOARD

Milk-Based Coffee Machine



Consept Style

The concept focuses on creating a compact and user friendly coffee machine for home use. The main idea is to make preparing coffee as easy as taking a drink from the refrigerator. The interface is designed to be simple and intuitive allowing users to quickly prepare milk based coffee with minimal effort while keeping a clean and minimal design language.

Function Analysis

The machine is designed to prepare different milk based coffee drinks through simple selection buttons. Each button represents a different drink option allowing users to prepare coffee quickly without complicated adjustments.

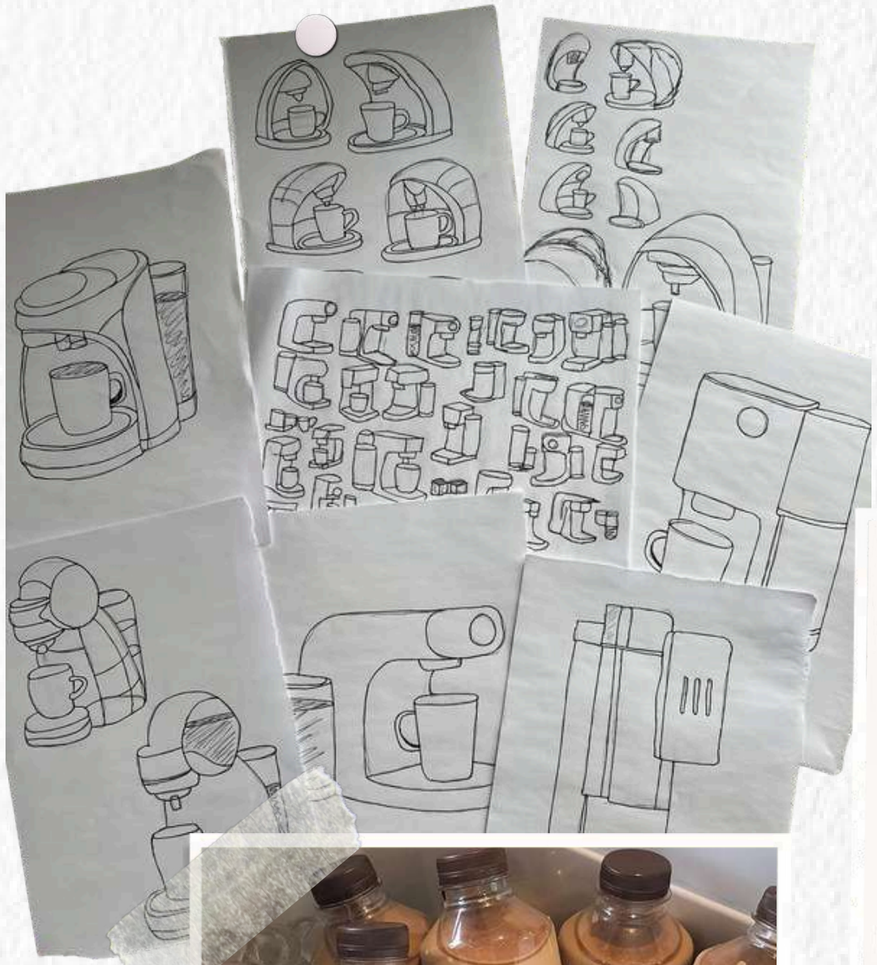
User Analysis

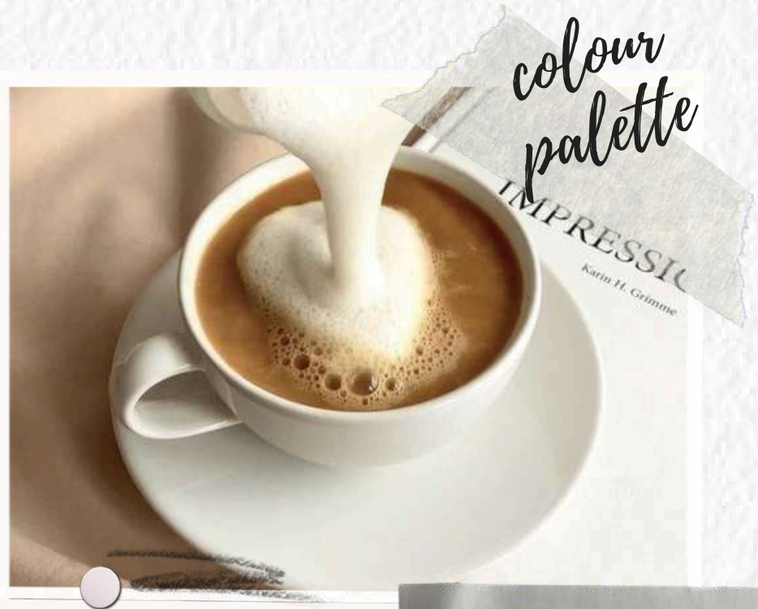
The product is designed for people who want to prepare milk-based coffee quickly and easily at home. Many users prefer simple machines that do not require complicated settings. Therefore i designed this product which focuses on an intuitive interface and a compact structure suitable for daily kitchen use.

Location Analysis

The product is intended for home kitchens especially for users who have limited counter space like small houses and kitchens. Its compact form allows it to fit easily into small kitchen environments while still providing a full coffee preparation experience.

MOODBOARD





colour palette

Compact, easy and fast!

FINAL PRODUCT

Colour Options

Materials

The coffee machine body is made of matte **ABS plastic** to provide durability and a clean minimal appearance. The water and milk reservoirs are produced from **transparent tritan plastic** so users can easily see the liquid level. The coffee outlet nozzle is **stainless steel** to ensure hygiene and heat resistance while internal parts such as tubes are made from **silicone** for safe liquid transfer.

Exploded View

Render of the Product

Why does it look like this?

The form focuses on simplicity and clarity. The cubic front surface creates a clear interaction area where the user can easily see and access the coffee selection buttons. The compact vertical body minimizes counter space while integrating water and milk containers.

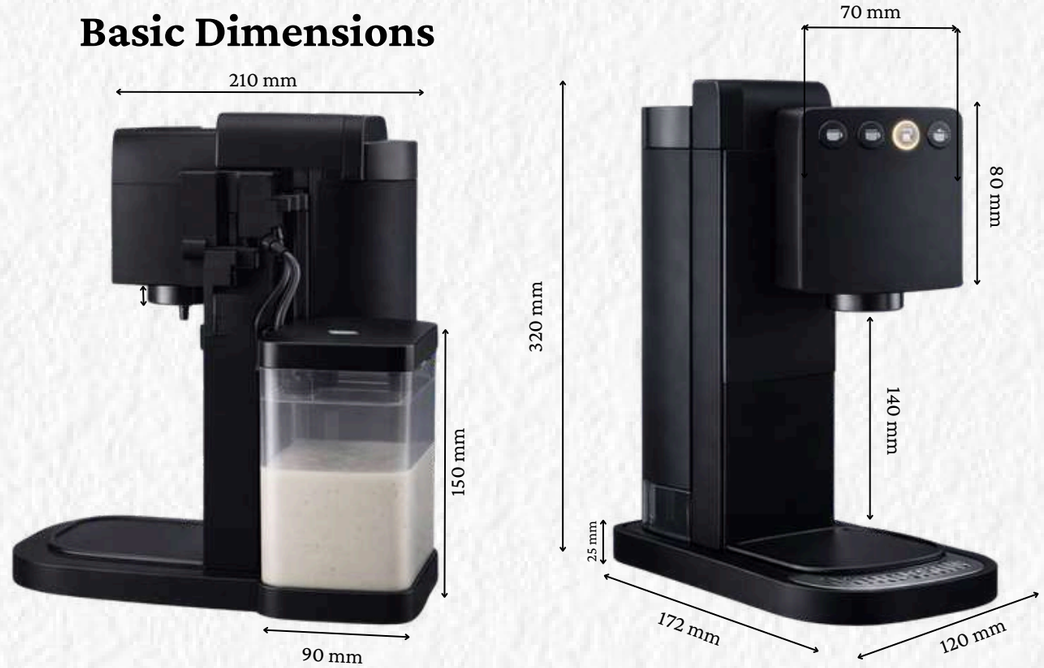
User Scenario

Product Features

- One-touch milk coffee preparation
- Compact footprint for small kitchens
- Dedicated drink selection buttons
- Transparent water reservoir
- Easy cleaning drip tray

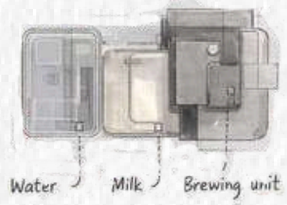
Detail View

Basic Dimensions



Cross Section

Top View



Simple Mechanism Diagram